INCH-POUND

MIL-DTL-17/97D
w/AMENDMENT 1
06 February 2026
SUPERSEDING
MIL-DTL-17/97D
19 June 2015

MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 93 OHMS, M17/97-RG210

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

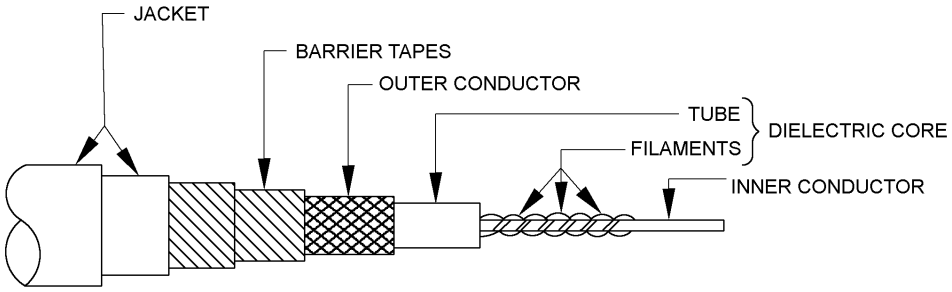


FIGURE 1. Configuration.

TABLE I. Description.

Components	Construction details
Inner conductor	Solid, silver-coated, copper-covered steel wire. Diameter: .0253 inch ± .0010.
Dielectric core	Type F-4: Air-spaced PTFE. A braid of four filament threads .028 inch approximate diameter each, with 2 picks/inch approximate, under two wraps, .005 inch thick each, with a 50% minimum overlap. Diameter: .146 inch ± .005.
Outer conductor	Single braid of AWG No. 34, silver-coated copper wire. Diameter: .191 inch, maximum. Coverage: 94.3%, nominal (Alternate 96.5% nominal) Carriers: 16 (Alternate 24) Ends: 7 (Alternate 5) Picks/inch: 8.2 ± 10% (Alternate 12.3 ± 10%)
Barrier tapes	Type FF-2: Two wraps of PTFE tape, .005-inch thick each, by 1/2-inch wide, with a 50% overlap, minimum.
Jacket	Type V: Double braid of fiberglass. Diameter: .242 inch ± .008.

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ENGINEERING INFORMATION:

Continuous working voltage: 750 V rms, maximum.
Operating frequency: 3 GHz, maximum.
Velocity of propagation: 88 percent, nominal.
Power rating: See figure 2.
Operating temperature range: -55° to +200°C.
Inner conductor properties:
DC resistance (maximum at 20°C): 4.4 ohms per 100 feet.
Elongation: 1 percent, minimum.
Tensile strength: 110 $klb_f/inch^2$, minimum.
Engineering note: This cable useful in low capacitance, high temperature applications (see connector series “TNC” and “BNC” per MIL-PRF-39012.)

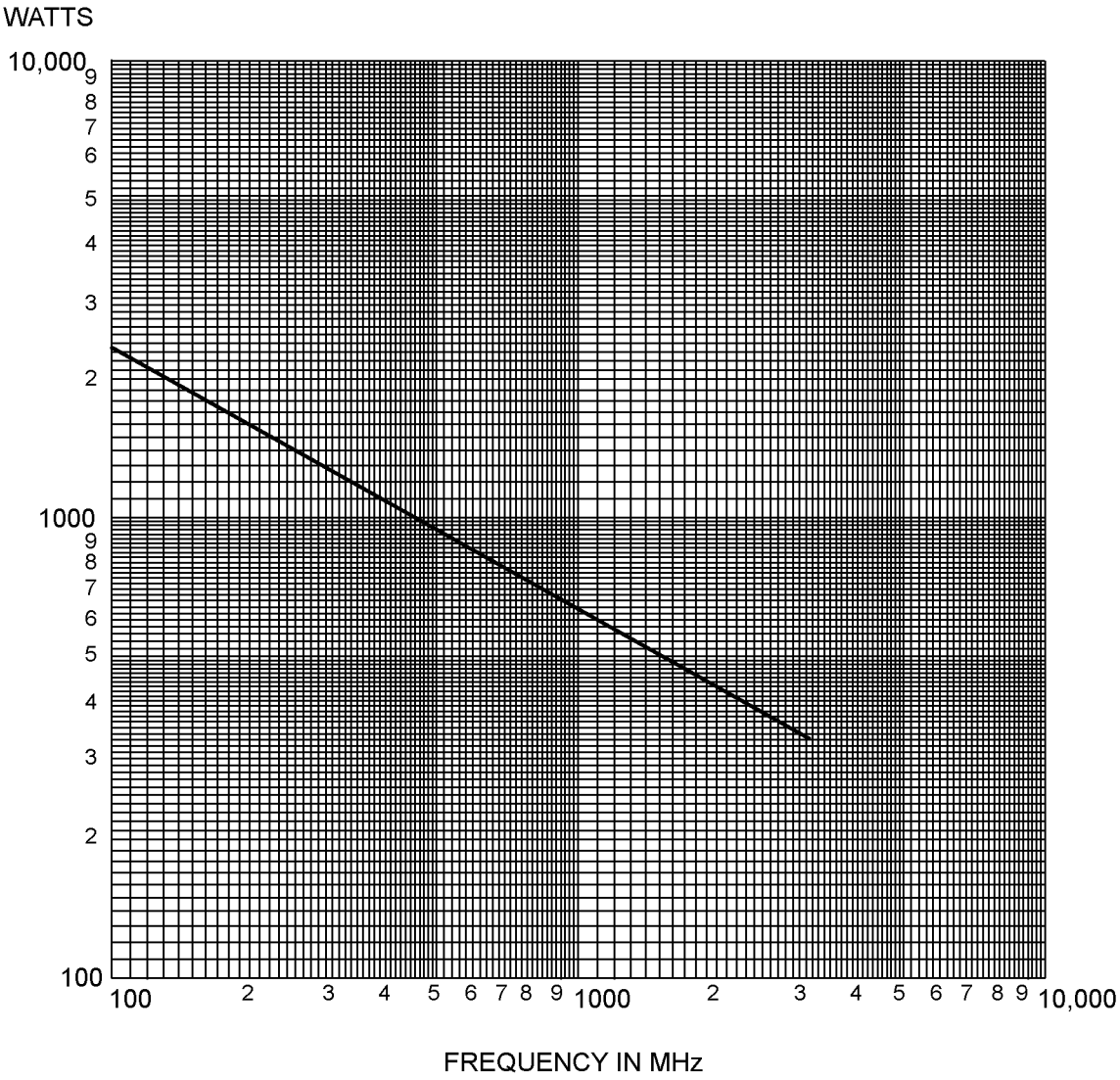
REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.
Environmental and mechanical:
Visual and mechanical examination:
Out-of-roundness: Not applicable.
Eccentricity: Not applicable.
Adhesion of conductors: Not applicable.
Aging stability: +250°C ± 5°C.
Stress-crack resistance: Not applicable.
Outer conductor integrity: Not applicable.
Cold bend: -55°C ± 2°C.
Dimensional stability: Not applicable.
Contamination: Not applicable.
Bendability: Not applicable.
Flammability: Not applicable.
Weight: 5 pounds per 100 feet, maximum.

Electrical:

Continuity: Applicable.
Spark test: Not applicable.
Voltage withstanding: 3,000 V rms, +10%, -0%.
Insulation resistance: Not applicable.
Corona extinction voltage: Not applicable.
Characteristic impedance: 93 ohms ± 5.
Attenuation: 8 dB per 100 feet, maximum, at 400 MHz.
Structural return loss: Not applicable.
Capacitance: 14.5 pF per foot, maximum.
Capacitance stability: ± 3 percent.
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Mechanically induced noise voltage: Not applicable.
Time delay: Not applicable.

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Power	
MHz	Watts
100	2400
400	1000
1000	630
2000	420
3000	340

FIGURE 2. Power rating at 25°C at sea level.

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Part or Identifying Number (PIN): M17/97-RG210.
Marking: Applicable
Marking Durability: 125 cycles (250 strokes), 1 pound
Workmanship: Applicable
Qualification: Shall consist of the inspections specified in table III.

Supersession data: See table II.

TABLE II. Cross reference of part number.

PIN	Superseded PIN or type designation
M17/97-RG210	RG210/U

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TABLE III. Qualification and Group A, B, and C Table.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
I	In-process inspection:		4.5	Entire sample		
	Continuity	3.7.1	4.8.2	Entire sample		
	Spark test	N/A				
	Voltage withstanding	3.7.3	4.8.4	Entire sample		
	Insulation resistance	N/A				
	Visual and mechanical inspection	3.6	4.8.1	Entire sample	Per Table VIII	A
	Diameter measurements.	3.6.1	4.8.1.1	Entire sample	Per Table VIII	A
	Out-of-roundness	N/A				
	Eccentricity of inner conductor	N/A				
	Adhesion of conductors	N/A				
	Marking	3.8	4.8.1	Entire sample	Per Table VIII	A
	Marking Durability	3.8.2	4.8.39	3	Per Table VIII	A
	Workmanship	3.10	4.8.1	Entire sample	Per Table VIII	A
II	Corona extinction voltage	N/A				
	Characteristic impedance	3.7.6	4.8.7	1	Per Table VIII	A
	RF transmission loss (attenuation)	3.7.7	4.8.8	2	Per Table VIII	A
	Standing wave ratio (return loss)	N/A				
	Capacitance	3.7.9	4.8.10	1	Per Table VIII	B
	Capacitance stability	3.7.10	4.8.11	1	Per Table VIII	C
	Capacitance unbalance	N/A				
	Transmission unbalance	N/A				
	Inductance	N/A				
	Mechanically induced noise voltage	N/A				
	Time delay	N/A				
	Aging stability	3.7.15	4.8.16	4	Per Table VIII	C

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TABLE III. Qualification and Group A, B, and C Table – Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
II	Stress crack resistance	N/A				
	Outer conductor integrity	N/A				
	Cold bend	3.7.18	4.8.19	3	Per Table VIII	B
	Dimensional stability	N/A				
	Contamination	N/A				
	Bendability	N/A				
	Flammability	N/A				
	Flame propagation	N/A				
	Acid gas generation	N/A				
	Halogen content	N/A				
	Immersion	N/A				
	Smoke index	N/A				
	Toxicity index	N/A				
	Durometer hardness	N/A				
	Weathering	N/A				
	Abrasion resistance	N/A				
	Tear strength	N/A				
	Heat distortion	N/A				
	Physicals	N/A				
	Tensile strength and elongation	N/A				
	Weight	3.9	4.8.24	1	Per Table VIII	B

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-17, this document references the following:

MIL-PRF-39012

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2026-007)

Review activities:
Army – AT, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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